

Learning to take a deep breath—with BDNF

George B Richerson & John M Bekkers

Short bursts of hypoxia induce a form of learning in the neuronal network that controls breathing. This process now is shown to share mechanisms with synaptic plasticity in the hippocampus.

Some synapses in the brain undergo long-term changes in their strength in response to activity. This synaptic plasticity is widely studied in the hippocampus, a part of the brain linked to memory, because the phenomenon is believed to contribute to memory. If that is the case, then the last place you would expect to find synaptic plasticity is in the neural network responsible for breathing. But the respiratory system does 'learn'; that is, its output is altered in response to prior experience. Just three short episodes of hypoxia can induce a form of respiratory memory called long-term facilitation (LTF).

In the January issue of *Nature Neuroscience*, Baker-Herman *et al.*¹ show that respiratory LTF shares mechanisms with synaptic plasticity in the hippocampus. The findings imply that it might be possible to 'teach' people to learn how to breathe better, particularly if they have one of the many disorders that can cause problems with breathing, such as cervical spinal cord injury, sleep apnea, neuromuscular disease or emphysema.

There are different forms of plasticity in the respiratory system², depending on the type of exposure to hypoxia (continuous versus intermittent, for example). Of these adaptations, LTF has been particularly well studied, in part because the short, repeated episodes of hypoxia used to induce LTF in animals are similar to the hypoxic episodes that occur during sleep in people with sleep apnea. The episodes also occur in normal people exercising at high altitudes, such as skiers who undergo repeated short episodes of pounding moguls separated by periods of rest.

LTF is expressed after return to normoxia³ as a long-term increase in breathing. This is usually measured by recording the output of the phrenic nerve, which goes to the diaphragm

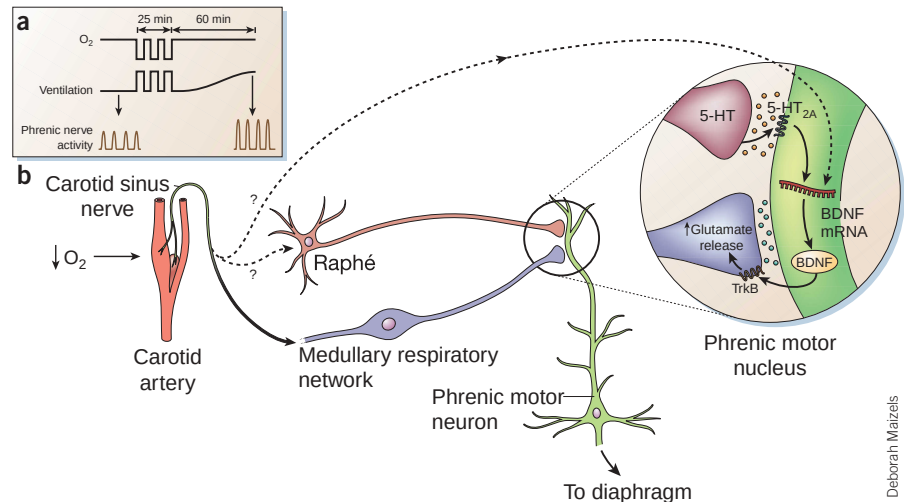


Figure 1 Respiratory memory through long-term facilitation (LTF). **(a)** When the oxygen level in inhaled air is lowered, the carotid body induces an immediate and transient increase in ventilation by stimulating the medullary respiratory network. After three short bursts of hypoxia, a delayed increase in ventilation and phrenic nerve activity occurs (shown as 'integrated' activity). Increased phrenic nerve activity results in deeper breaths, beginning around 25 minutes after return to normoxia, and lasts for at least an hour. **(b)** Baker-Herman *et al.* report that this 'learning' behavior (LTF) is caused by an increase in BDNF synthesis in the cervical spinal cord, which in turn stimulates nearby TrkB receptors. The result is an increase in firing of phrenic motor neurons in response to glutamatergic input from the medulla. The increase in BDNF and the expression of LTF both depend on serotonergic input from neurons in the medullary raphé nuclei. It is not yet clear whether hypoxic stimulation of the carotid body induces the increase in BDNF through the raphé nuclei or some other pathway. Shown are the locations of BDNF, TrkB and serotonin (5-HT) receptors favored by Baker-Herman *et al.*

and causes the lungs to inflate. LTF leads to a prolonged increase in lung ventilation that counteracts any future episodes of hypoxia (Fig. 1).

For intermittent hypoxia to induce LTF, the carotid body must be intact. The carotid body, located on the carotid artery in the neck, is a site of oxygen sensors. LTF can also be induced when the carotid body is bypassed by electrically stimulating the carotid sinus nerve, which travels from the carotid body to the brainstem. This observation indicates that LTF results from changes in the CNS, rather than from an increase in hypoxic sensitivity of the carotid body⁴. Earlier work showed that the LTF seen in phrenic nerve output could be blocked by applying serotonin receptor antagonists or a protein synthesis inhibitor to the cervical spinal cord. Beyond that, however, it has been difficult to define the central mechanisms of respiratory LTF.

Baker-Herman *et al.*¹ have now developed a clever approach using an *in vivo* rat preparation to study the mechanisms of LTF. The investigators set out specifically to examine the role of brain-derived neurotrophic factor (BDNF) in LTF. BDNF can be secreted from neurons in response to increased activity, and is required for some forms of hippocampal long-term potentiation (LTP), an increase in the size of excitatory postsynaptic potentials induced by patterned stimulation.

LTP and LTF are both forms of learning, but LTF is a change in the output of the brain that is clearly linked to a clinically relevant behavior, whereas LTP is the plasticity of synapses in the brain whose functional significance remains a matter of conjecture. Early work on LTP suggested that BDNF production occurs in postsynaptic dendrites and acts as a retrograde messenger that activates TrkB receptors on presynaptic terminals to enhance glutamate release⁵. Later work showed that not all forms

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of LTP require BDNF, and that when BDNF is required there are different ways it can act—it can sometimes be produced by the presynaptic neuron, and can work either presynaptically or postsynaptically^{6,7}. The specific role of BDNF may thus depend on the synapse being studied and the protocol used to induce LTP.

To determine whether there are parallels between the mechanisms of respiratory LTF and hippocampal LTP, Baker-Herman *et al.* examined BDNF levels in the cervical spinal cord in response to intermittent hypoxia (inhaling 11% O₂ three times for five minutes each, every ten minutes). The authors found that BDNF increased in proportion to the magnitude of LTF. The increase in BDNF was blocked by manipulations that inhibited LTF, including transection of the carotid sinus nerve and application of a serotonin receptor antagonist or protein synthesis inhibitor to the spinal cord. The authors next showed that application of BDNF to the spinal cord led to an increase in phrenic nerve output that was similar to the LTF induced by intermittent hypoxia.

These results all pointed to an association between BDNF production and LTF, but did not show that BDNF was necessary for LTF. To demonstrate this directly, the authors blocked BDNF with BDNF-specific small interfering RNA (siRNA). After application of siRNA to the spinal cord, intermittent hypoxia no longer led to LTF or to an increase in BDNF. Pharmacological blockade of TrkB receptors also prevented LTF.

The authors propose the following working model for LTF (Fig. 1). Intermittent hypoxia induces an increase in carotid sinus nerve activity, which leads to activation of phrenic motor neurons. When combined with release of serotonin into the cervical spinal cord (at baseline levels, or perhaps at levels increased by hypoxia), BDNF is synthesized in dendrites of phrenic motor neurons. BDNF then acts on TrkB receptors on presynaptic terminals from medullary respiratory neurons to enhance the excitatory synaptic input to phrenic motor neurons.

Future work will be needed to determine how closely the mechanisms of respiratory LTF parallel those of hippocampal LTP. For respiratory LTF, the specific cellular locations of serotonin receptors, BDNF production and TrkB receptors remain unclear. It has also not been shown directly that enhancement of phrenic output is caused by an increase in strength of the synapses from medullary respiratory neurons and, if so, whether this is induced by high-frequency stimulation of these synapses, an increase in serotonin release, or a combination of the two. Some of

these questions will be very difficult to address *in vivo* and may require the development of *in vitro* models of respiratory LTF.

Many human conditions that lead to abnormalities of breathing are often fatal or severely incapacitating. Patients with cervical spinal cord injuries or neuromuscular diseases, for example, often require mechanical ventilation. Patients who have retained some ability to contract their diaphragm would greatly benefit if the output of the phrenic nerve could be enhanced. The current results suggest that BDNF and serotonin may become important therapeutic targets, but it is now important to determine how closely plasticity in the human respiratory system parallels that seen in this

animal model of LTF. A clear definition of the mechanisms of respiratory LTF offers the hope of harnessing the capacity of the respiratory network to remember, so that patients suffering from breathing disorders might someday learn how to enhance their respiratory output.

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Androgen receptor outwits prostate cancer drugs

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Almost all prostate cancer patients become resistant to therapy that blocks androgen-mediated cell proliferation. The key to this resistance may lie in expression of the androgen receptor itself (pages 33–39).

Most patients with metastatic prostate cancer receive drugs that block production of androgen, which promotes cancer cell proliferation. These drugs rarely actually cure cancer, however, because the initial response is almost always followed by a relapse to an unresponsive, hormone-refractory stage. In an attempt to block this refractory state, competitive androgen receptor antagonists (antiandrogens) are often combined with drugs that block androgen production (mainly analogs of luteinizing hormone-releasing hormone, LHRH). Despite such combination treatment, one man will die of hormone-refractory prostate cancer every 20 minutes in the United States alone. These sobering facts have led to extensive studies to define the molecular changes required for the refractory state.

In this issue, Chen *et al.* place the androgen receptor at the center of this process¹. The authors report that a two- to fivefold increase in androgen receptor mRNA is the only gene

expression change consistently associated with hormone-refractory disease. As a result, cells become supersensitive to androgen rather than independent of it, and antiandrogens can even end up promoting androgen function instead of inhibiting it. The study sets the stage for the development of therapies that can prevent progression to the refractory stage.

In the normal prostate, androgen action is a paracrine process with different effects on different types of cells. Binding of androgen receptor by its cognate ligand results in dimerization and binding of the receptor to regulatory sequences of specific genes in the nuclei of epithelial and stromal cells (Fig. 1). In the stromal cells, androgen receptor activates secretion of peptide growth factors known as andromedins². These andromedins diffuse across the basement membrane into the epithelial compartment, which contains both basal and secretory luminal cells. There, the andromedins bind to specific plasma membrane receptors to initiate signaling for proliferation and epithelial cell survival. In the nuclei of the secretory cells, androgen receptor stimulates the transcription of a series of genes encoding prostate-specific differentiation markers, including prostate-specific antigen (PSA) and human glandular kallikrein-2. Although androgen receptor transcriptionally regulates these markers in the

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